

SHUBHAM MOOKIM

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EXPERIENCE

• ForGood.ai [🌐]

Software Developer — AI Systems

January 2025 – Present

Bengaluru, India

- Architected **Patient OS v2** — production multi-agent AI copilot for cancer patients; context budget system (60K-char across 6 sources), Redis conversation compression (140x latency speedup), Apple Health (30+ metrics), GPT-4o vision, and parallel medical tool execution (PubMed, CIViC, DrugBank, clinical trials)
- Built **FairEdge Data Agent** — NL-to-SQL agentic pipeline (Agno + LangChain + AWS Athena); multi-agent flow with guardrails, session management, dynamic column-fetch, and multi-modal output; sub-2s latency, deployed to live enterprise HR analytics client
- Designed **QC Agent System** — 3-agent pipeline: Scanner generates per-file JSON metadata from React/React Native codebases; Generator produces unit, UI, functional & API test cases via LLM; Verification Agent validates correctness before output
- Built **GeneSilico OCR Service** (Gemini Vision + Mistral AI + FastAPI) for medical TRF document extraction; **Medical Transcription Service** on AWS SAM with Lambda, Kinesis Firehose audio pipeline
- Owned 8+ pHow it works is we have one section called Case books, in which, on the left pane, we have some sort of a sidebar where you can select one patient. The user was logged in and can only see the transcription. The thing is, the oncologist can see it, so you can get the email ID of the user from there, in which the thing will be handled by the oncologist and reduction services (Node.js APIs, Next.js, React Native iOS/Android); led HIPAA compliance — PHI redaction, encryption, IAM hardening; set up full observability stack (PostHog, DataDog RUM, Prometheus/Grafana, Arize Phoenix)

• BrinxAI (Remote)

AI Infrastructure Security Engineer (Freelance)

October 2024 – December 2024

Cyprus-based AI Infrastructure Company

- Deployed OpenVPN to secure distributed AI compute nodes; hardened node access controls and remediated infrastructure vulnerabilities

• Houzee India [🌐]

Mobile Development Intern

July 2024 – October 2024

Bengaluru, India

- Built cross-platform mobile app screens using React Native and TypeScript; developed Express.js backend integrations for real-time data features

• STARC, PES University [🌐]

ML Research Intern

April 2023 – June 2023

Bengaluru, India

- Built a computer vision ML pipeline for cricket sports analytics — automated ball tracking, batsman detection, and per-ball event extraction from match footage using Google ML tools and OpenCV

PROJECTS

• Patient OS v2 — Multi-Source Health AI Copilot

2025

Stack: Python, Agno, OpenAI GPT-4o, FastAPI, MongoDB, Redis Cluster, AWS, SSE Streaming

- Production oncology AI agent that reasons simultaneously over clinical genomics, uploaded medical records, Apple Health wearable data (30+ metrics), and live medical literature
- **ContextBudgetManager** allocates a 60K-char context window across 6 prioritised sources; Redis-cached conversation compression delivers 3ms cache hits vs 420ms fresh — a 140x speedup
- **GPT-4o Vision** for medical image and food photo analysis; parallel tool executor runs PubMed, CIViC, DrugBank, and clinical trials lookups simultaneously in a single response pass

• FairEdge Data Agent — Enterprise NL-to-SQL Agentic Pipeline

2025

Stack: Python, Agno, LangChain, AWS Athena, FastAPI, Redis, Docker

- Built a multi-agent system that lets non-technical teams query large structured datasets using plain English — no SQL knowledge required; deployed to a live enterprise HR analytics client
- Queries pass through guardrails, a session manager, and a classifier before the agent fetches relevant column metadata and generates optimised SQL, executed serverlessly on AWS Athena; responses include both natural language summaries and rendered graphs

• AI-Powered QC Agent System — Automated Test Generation

2025

Stack: Python, LLM APIs, React/React Native AST Analysis, FastAPI

- A Scanner Agent analyses React and React Native codebases file-by-file, producing structured JSON metadata; a Generator Agent uses this to write unit, UI, functional, and API test cases via LLM; a Verification Agent checks every output for syntax and semantic correctness before acceptance

- Eliminated manual test-case writing from the QC workflow — test coverage now generated automatically on any codebase
- **Medical Transcription Service — Serverless Healthcare Backend** 2025
Stack: AWS SAM, Lambda, Kinesis Firehose, S3, API Gateway, CloudFormation, Python
 - Doctors record or upload patient sessions; the system automatically transcribes audio and generates structured medical reports via NeuroGPT integration — zero manual effort post-recording
 - Serverless architecture using 8 independently deployable CloudFormation stacks; Kinesis Firehose handles real-time audio chunking, transcription triggering, and stale session cleanup via dedicated Lambda functions

SKILLS

- **AI / LLM:** Agno, LangChain, LangGraph, LlamaIndex, OpenAI API (GPT-4o), Gemini Vision API, Mistral AI, RAG, Multi-Agent Systems, Prompt Engineering, Arize Phoenix, LangSmith, PostHog
- **Backend & APIs:** Python (FastAPI, Celery), Node.js (Express.js), REST APIs, SSE Streaming, JWT Auth, Microservices
- **Cloud & Infrastructure:** AWS (Lambda, SAM, Kinesis Firehose, S3, Athena, API Gateway, EC2, ECR, CloudFormation), Docker, Kubernetes, GitHub Actions, OpenVPN
- **Databases:** MongoDB, Redis (Cluster, Replication), PostgreSQL, AWS Athena, SQL, DynamoDB
- **Frontend & Mobile:** React (Next.js), React Native (iOS + Android), TypeScript, Tailwind CSS, Redux Toolkit, React Query, Firebase
- **Observability & Security:** Prometheus, Grafana, DataDog RUM, AWS CloudWatch, HIPAA Compliance, PHI Redaction, IAM Policy Design
- **Languages:** Python, TypeScript, JavaScript, SQL, Bash, C++
- **Core Concepts:** System Design, Distributed Systems, Serverless Architecture, Event-Driven Architecture, Data Structures & Algorithms

EDUCATION

- **PES University** 2021 – 2025
B.Tech — Electronics & Communications Engineering (Major) + Computer Science (Minor) Bengaluru, India
- **National Centre For Excellence** 2018 – 2021
Secondary Education (CBSE — 92.3%) Bengaluru, India

HONORS & AWARDS

- **Merit Scholarship** 2019
National Centre For Excellence 
 - Awarded scholarship for securing 92.3% in CBSE Board Examinations
- **Top 5 Teams — Qubitrix Hackathon** 2023

 - Team placed in top 5 out of 30 competing teams

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Fluent), Kannada (Basic) **Interests:** AI Systems, Health Tech, Open Source, Guitar, Sports